SRI KRISHNA COLLEGE OF TECHNOLOGY
(An Autonomous Institution)
Affiliated to Anna University and Approved by AICTE
KOVAIPUDUR, COIMBATORE – 641 042.

SRI KRISHNA INSTITUTIONS

FLEET MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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In partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

INFORMATION TECHNOLOGY

SRI KRISHNA COLLEGE OF TECHNOLOGY

Autonomous Institution | Approved by AICTE | Affiliated to Anna University | Accredited by NAAC with A++ Grade

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SUSTAINABLE DEVELOPMENT GOALS

The Sustainable Development Goals are a collection of 17 global goals designed as a blueprint to achieve a better and more sustainable future for all. The SDGs were set in 2015 by the United Nations General Assembly and are intended to be achieved by the year 2030. The Fleet Management System contributes to sustainable transportation, efficient resource utilization, and improved fleet operations. It promotes smarter fleet planning while reducing operational costs and environmental impact. It also supports safer, more reliable, and data-driven transportation management.

Questions	Answer Samples
Which SDGs does the project directly address?	**SDG 8, SDG 11 and SDG 12** – promoting efficient work, sustainable transportation, and responsible resource usage.
What strategies or actions are being implemented to achieve these goals?	Vehicle utilization, fuel monitoring, preventive maintenance, driver monitoring, and trip management.
How is progress measured and reported in relation to the SDGs?	By tracking fuel usage, vehicle utilization, maintenance costs, and vehicle downtime.
How were these goals identified as relevant to the project's objectives?	They support efficient transportation, reduce resource wastage, and improve fleet productivity.
Are there any partnerships or collaborations in place to enhance this impact?	Collaboration with fleet operators, service centers, fuel providers, and regulatory authorities can enhance the system's impact.

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BONAFIDE CERTIFICATE

Certified that this project report titled **" FLEET MANAGEMENT SYSTEM "** is the Bonafide work of **DIVYA DHARSHINI M (727824TUIT043)** who carried out the project work under my supervision.

SIGNATURE **SIGNATURE**

Submitted for the Project viva-voce examination held on _____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

ACKNOWLEDGEMENT

At this juncture, we take the opportunity to convey our sincere thanks and gratitude to the management of the college for providing all the facilities to us.

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ABSTRACT

The Fleet Management System is a web-based application developed to provide an integrated and efficient solution for managing fleet operations. Traditional fleet management often involves separate records and manual processes for handling vehicles, drivers, trips, fuel consumption, maintenance, and compliance documents, which can lead to delays, errors, and increased operational costs. The proposed system brings these activities together through a centralized platform.

The system enables authorized users to register and manage vehicles, maintain driver information, assign trips, monitor vehicle and driver performance, schedule preventive maintenance, track fuel usage, and manage important documents such as registration, insurance, pollution, and fitness certificates. Role-based access control ensures that administrators, fleet managers, drivers, maintenance personnel, and finance users can access functions according to their responsibilities.

The system also provides notifications for document expiry and maintenance requirements, along with analytical reports for monitoring fleet utilization, fuel efficiency, maintenance costs, and driver performance. By providing centralized data and timely insights, the system helps reduce manual effort, improve vehicle availability, minimize operational expenses, and support better decision-making. Overall, the Fleet Management System promotes safer, more reliable, cost-effective, and sustainable fleet operations.

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LIST OF ABBREVIATIONS

Sl.No	Abbreviation	Expansion
1.	API	Application Programming Interface
2.	DFD	Data Flow Diagram
3.	DOM	Document Object Model
4.	ERP	Enterprise Resource Planning
5.	GPS	Global Positioning System
6.	HTTP	Hypertext Transfer Protocol
7.	JDBC	Java Database Connectivity
8.	JPA	Java Persistence API
9.	JSON	JavaScript Object Notation
10.	JSX	JavaScript XML
11.	JWT	JSON Web Token
12.	OBD	On-Board Diagnostics
13.	OS	Operating System
14.	RAM	Random Access Memory
15.	REST	Representational State Transfer
16.	SQL	Structured Query Language
17.	UI	User Interface
18.	UML	Unified Modeling Language

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The rapid growth of transportation and logistics operations has increased the need for efficient and organized fleet management. Traditional fleet management methods often rely on manual records and separate processes for handling vehicles, drivers, trips, fuel, maintenance, and compliance documents. These challenges can lead to operational delays, increased costs, poor vehicle utilization, and difficulties in maintaining accurate fleet information.The Fleet Management System aims to address these issues by providing a centralized, user-friendly platform for managing fleet operations. The system simplifies vehicle and driver management, trip assignment, maintenance scheduling, fuel monitoring, and document tracking, while enabling authorized users to efficiently monitor and manage fleet activities.

The platform allows fleet managers to maintain vehicle records, assign drivers and trips, monitor fuel usage, schedule preventive maintenance, and track important documents such as registration, insurance, pollution, and fitness certificates. At the same time, drivers and other authorized personnel can access the information and functions relevant to their assigned responsibilities.One of the key benefits of the system is its ability to reduce manual work, provide timely notifications, improve data accuracy, and support better utilization of fleet resources. By integrating major fleet operations into a single platform, the system improves transparency and enables faster access to important information for effective decision-making.

Ultimately, the Fleet Management System aims to streamline fleet operations, reduce operational costs, improve vehicle reliability and utilization, and provide a modern and efficient approach to fleet management.

1.2 COMPONENTS OF SYSTEM

System Landing Page

The landing page serves as the entry point for users, providing an overview of the fleet management services and system features. It allows users to access login and registration and navigate to the appropriate modules based on their roles.

System Dashboard

The dashboard acts as the central hub for fleet operations. Fleet managers can view vehicle availability, active trips, maintenance status, fuel usage, and important alerts, while other authorized users can access information related to their responsibilities.

Vehicle Management

This component allows authorized users to register, update, and manage vehicle information such as vehicle number, type, model, status, and assigned driver. It provides a centralized record of all vehicles in the fleet.

Driver Management

The driver management component maintains driver profiles, license details, assigned vehicles, trip information, and performance records. It helps fleet managers assign suitable drivers and maintain accurate driver information.

Trip and Route Management

This component enables users to create and manage trips by assigning vehicles and drivers, defining routes, and recording trip details. It helps organize transportation activities and monitor planned and completed trips.

Maintenance Management

The maintenance component supports preventive maintenance scheduling and maintenance record management. It helps identify upcoming service requirements and reduces vehicle downtime through timely maintenance.

Fuel Management

This component records and monitors fuel consumption and related expenses for fleet vehicles. It helps identify fuel usage patterns, calculate fuel efficiency, and control unnecessary fuel expenditure.

Document and Compliance Management

This component manages important vehicle and driver documents such as registration, insurance, pollution certificates, fitness certificates, and driving licenses. It provides alerts for upcoming expirations to support timely renewal and compliance.

Driver Behaviour Monitoring

The system records relevant driver performance information such as speeding, harsh braking, excessive idling, and other driving events. This information can be used to generate driver performance scores and encourage safer driving practices.

Notifications and Alerts

The notification component provides timely alerts for document expiry, scheduled maintenance, trip updates, and other important fleet activities. This helps users take necessary actions without delay.

Analytics and Reporting

This component provides reports and insights on vehicle utilization, fuel consumption, maintenance costs, trip performance, and driver performance. It helps fleet managers make informed operational decisions.

User Profile Management

Users can manage their personal information and access role-specific features through their profiles. Role-based access ensures that users can only access the functions permitted for their responsibilities.

Security and Access Control

The system uses secure authentication and role-based access control to protect fleet and user information. Only authorized users can access sensitive data and perform management operations.

1.3 ADVANCED TECHNOLOGIES

Intelligent Fleet Optimization

AI-based insights help fleet managers optimize vehicle utilization, trip assignments, and operational planning based on fleet data and usage patterns.

Predictive Maintenance

Maintenance data can be analyzed to identify upcoming service requirements and potential vehicle issues, helping reduce unexpected breakdowns and vehicle downtime.

Fuel Efficiency Monitoring

The system monitors fuel consumption and efficiency for individual vehicles, helping identify excessive fuel usage and reduce unnecessary operational costs.

Real-Time Notifications

Automated notifications alert users about trip updates, maintenance schedules, document expirations, and other important fleet activities, improving timely decision-making.

Driver Performance Monitoring

Driver activities such as speeding, harsh braking, and excessive idling can be monitored to generate performance insights and encourage safer driving practices.

Mobile-Friendly Platform

A responsive interface allows drivers and fleet personnel to access trip details, vehicle information, notifications, and other essential functions conveniently from mobile devices.

Data Analytics for Fleet Managers

Analytical dashboards provide insights into vehicle utilization, fuel consumption, maintenance costs, trip performance, and driver performance, helping managers make informed decisions.

1.4 GLOBAL PERSPECTIVES

Global perspectives on fleet management systems are shaped by the growth of transportation and logistics, technological advancements, rising operational costs, and the need for efficient vehicle utilization.

• Growing Demand for Efficient Fleet Operations

With the expansion of transportation and logistics activities worldwide, organizations require efficient systems to manage vehicles, drivers, trips, and operational resources effectively.

• Shift Toward Digital Fleet Management

Organizations are increasingly moving from manual records to digital fleet management platforms. Web and mobile-based systems provide centralized access to vehicle information, trip details, maintenance records, and compliance data.

• Integration of Advanced Technologies

Artificial intelligence, data analytics, IoT, and connected vehicle technologies are

being integrated into fleet management to improve maintenance planning, fuel efficiency, driver performance, and vehicle utilization.

- **Focus on Safety and Compliance**

Modern fleet systems emphasize driver safety, vehicle maintenance, and timely renewal of essential documents. Automated alerts and performance monitoring help organizations maintain safer and more compliant fleet operations.

- **Data-Driven Fleet Optimization**

Globally, fleet management platforms are evolving to provide analytical insights into fuel consumption, maintenance costs, vehicle utilization, and driver performance. This helps organizations reduce operational expenses and make informed decisions.

CHAPTER – 2

SYSTEM ANALYSIS

2.1 EXISTING SYSTEM

• Many organizations currently manage fleet operations using manual records, spreadsheets, phone calls, and separate software tools for vehicles, drivers, trips, fuel, and maintenance. This can lead to data duplication, errors, and difficulty in accessing updated information.

• Existing digital fleet systems may provide basic vehicle and driver management, but often have limited integration between trip planning, maintenance, fuel monitoring, and compliance management.

• Maintenance activities are frequently handled based on fixed schedules or manual reminders. This can result in delayed servicing, unexpected vehicle breakdowns, and increased maintenance costs.

• Fuel consumption and driver performance are often recorded separately, making it difficult for fleet managers to identify excessive fuel usage, inefficient driving behaviour, or underutilized vehicles.

• Document management is also a challenge when vehicle registration, insurance, pollution, fitness, and driver licence records are maintained manually. Missing renewal deadlines can create compliance issues.

• Overall, existing fleet management approaches may support basic operations but lack a centralized platform for monitoring fleet performance, reducing operational costs, improving vehicle utilization, and supporting data-driven decision-making.

2.1.1 DRAWBACKS

While existing digital and manual fleet management methods provide basic operational support, they also face certain challenges:

Quality of Service

• **Delayed Information:** Manual updates can result in outdated vehicle, driver, trip, and maintenance information.

• **Coordination Issues:** Separate processes for drivers, vehicles, maintenance, and trips can make communication and coordination difficult.

Limited Functionality

• **Lack of Integration:** Some systems do not combine vehicle, trip, fuel, maintenance, and compliance management in a single platform.

• **Limited Analytics:** Basic systems may provide insufficient insights into fuel efficiency, vehicle utilization, maintenance costs, and driver performance.

Operational Issues

• **Manual Errors:** Repeated manual data entry can lead to incorrect records and scheduling mistakes.

• **Vehicle Downtime:** Delayed maintenance can increase the possibility of breakdowns and reduce vehicle availability.

Security and Privacy

• **Data Security Risks:** Vehicle, driver, and operational information requires secure authentication and controlled access.

• **Unauthorized Access:** Without proper role-based permissions, sensitive fleet information may be accessible to unauthorized users.

Cost Considerations

• **Higher Operational Costs:** Poor fuel monitoring, inefficient vehicle utilization, and delayed maintenance can increase overall fleet expenses.

• **Resource Wastage:** Inefficient planning may result in unnecessary vehicle usage, fuel consumption, and maintenance expenditure.

2.2 PROBLEM DEFINITION

1. **Manual Management Issues:** Fleet information is often maintained through spreadsheets, paper records, and separate systems. This can result in data duplication, incorrect entries, difficulty in updating records, and delays in accessing important fleet information.
2. **Limited Monitoring:** Fleet managers may face difficulties in monitoring vehicle utilization, fuel consumption, maintenance status, trip activities, and driver performance from a single platform. This makes it harder to identify inefficiencies and respond to operational issues quickly.
3. **Maintenance and Compliance Delays:** Manual maintenance scheduling and document tracking can lead to missed service dates and document expiry deadlines. Delayed maintenance may increase vehicle downtime, while expired documents can create compliance and operational issues.
4. **Poor Decision-Making:** The absence of centralized data and analytical reports makes it difficult to evaluate fleet performance, identify unnecessary expenses, and recognize areas for improvement. Fleet managers may therefore depend on manual analysis for important operational decisions.

2.3 PROPOSED SYSTEM

• **Centralized Fleet Management:** The system provides a single platform to manage vehicles, drivers, trips, fuel, maintenance, and compliance records, reducing dependency on manual records and separate systems.

• **Vehicle and Driver Management:** Authorized users can add, update, and manage vehicle and driver details, including vehicle status, driver information, and relevant documents.

• **Trip and Route Management:** Fleet managers can create and assign trips to suitable vehicles and drivers, maintain trip details, and monitor planned and completed journeys.

• **Preventive Maintenance:** The system allows maintenance schedules to be planned and maintained, helping ensure timely servicing and reducing unexpected vehicle downtime.

• **Fuel Management:** Fuel consumption and expenses can be recorded and analyzed to monitor fuel efficiency and identify unnecessary fuel usage.

• **Document and Compliance Tracking:** The system maintains important vehicle and driver documents and provides timely alerts for upcoming document expirations and renewals.

• **Driver Performance Monitoring:** Driver-related performance information can be maintained to identify driving issues and support safer and more efficient fleet operations.

• **Analytics and Reporting:** The system provides reports on vehicle utilization, fuel consumption, maintenance costs, trips, and driver performance to support effective decision-making.

• **Notifications and Alerts:** Automated notifications help users stay informed about maintenance schedules, document expiry, trip updates, and other important fleet activities.

2.3.1 ADVANTAGES

• Centralized Fleet Management

• Improved Vehicle Utilization

• Reduced Manual Errors and Delays

• Better Maintenance and Compliance Management

• Improved Fuel Efficiency and Cost Control

• Better Communication and Coordination

• Time-Saving and Cost-Effective Operations

CHAPTER – 3

SYSTEM REQUIREMENTS

The hardware and software requirements and the platform description of the Fleet Management System are explained under Sections 3.1, 3.2, and 3.3 respectively.

3.1HARDWARE REQUIREMENTS

i. **Processor Type** : Intel Core i5
ii. **RAM** : 8 GB
iii. **Hard Disk** : 512 GB SSD
iv. **Display** : 1366 × 768 resolution or higher
v. **Internet Connection** : Stable broadband connection

3.2 SOFTWARE REQUIREMENTS

a. **Operating System** : Windows 11
b. **Front End** : React.js
c. **Back End** : Java with Spring Boot
d. **Database** : MySQL
e. **Development Environment** : Visual Studio Code
f. **Web Browser** : Google Chrome / Microsoft Edge

3.3 SOFTWARE DESCRIPTION



Fig . 3.1 VS Code Logo

Visual Studio Code, also commonly referred to as VS Code, is a source code editor made by Microsoft, it can be used to work with Windows, Linux and macOS. Features include support for debugging, syntax highlighting, intelligent code completion, snippets, embedded Git. Users can change the theme, keyboard shortcuts, preferences, and install extensions that add functionality.

3.3.1 FRONTEND

ReactJS

ReactJS is a JavaScript library used for developing interactive and responsive user interfaces for web applications. In the Fleet Management System, ReactJS is used to create dashboards and interfaces for managing vehicles, drivers, trips, maintenance, fuel, and compliance information. React follows a component-based architecture, allowing different parts of the application to be developed as reusable components.

React uses a Virtual DOM to efficiently update the user interface when application data changes. It also uses JSX, which allows developers to write HTML-like structures within JavaScript, making the interface easier to develop and maintain. React follows a unidirectional data flow, which provides a clear and predictable way of managing application data.

Features of ReactJS

1. **Component-Based Architecture**
   React divides the user interface into reusable components, making the application easier to develop, maintain, and update.
2. **Virtual DOM**
   The Virtual DOM improves application performance by updating only the required parts of the user interface.
3. **JSX**
   JSX allows developers to write HTML-like syntax within JavaScript, making UI components easier to understand.

4. **Reusable Components**
   Common elements such as navigation bars, dashboards, forms, tables, and cards can be reused across different modules.
5. **Unidirectional Data Flow**
   Data flows in a predictable direction between components, making application state easier to manage.
6. **Responsive Interface**
   React can be used to build interfaces that adapt to different screen sizes, allowing fleet personnel to access the system from desktops, tablets, and mobile devices.
7. **Rich Ecosystem**
   React provides access to a large ecosystem of libraries and development tools that support application development.

3.3.2 BACKEND

Java with Spring Boot

Java is a high-level, object-oriented programming language widely used for developing secure and scalable applications. In the Fleet Management System, Java is used to implement the backend logic and business operations. Spring Boot provides a framework for developing RESTful web services and connecting the frontend with the database.

The backend manages operations such as user authentication, vehicle management, driver management, trip assignment, maintenance records, fuel management, and document tracking. Spring Data JPA is used to simplify database operations, while Spring Security provides authentication and authorization mechanisms.

Extensions and Tools Used

1. **Extension Pack for Java**
   Provides essential Java development features such as code navigation, debugging, and IntelliSense.

2. **Spring Boot Extension Pack**
   Provides development support for creating and managing Spring Boot applications.
3. **Lombok**
   Reduces repetitive Java code by automatically generating getters, setters, constructors, and other commonly required methods.

Dependencies Used

1. **Spring Web**
   Provides the components required to create REST APIs and handle HTTP requests and responses.
2. **Spring Boot DevTools**
   Improves development efficiency through automatic application restart and development-time features.
3. **Spring Data JPA**
   Simplifies database access and enables interaction with database tables through JPA repositories.
4. **MySQL Driver**
   Provides connectivity between the Spring Boot application and the MySQL database.
5. **Lombok**
   Reduces boilerplate code and improves the readability and maintainability of Java classes.
6. **Spring Security**
   Provides authentication and authorization features to protect the system and control access based on user roles.
7. **JWT Authentication**
   JSON Web Tokens are used to securely authenticate users and maintain authorized access to protected backend resources.

CHAPTER – 4

SYSTEM DESIGN

4.1 MODULE DESCRIPTION

The Fleet Management System consists of the following five core modules:

1. **User & Driver Management**
2. **Vehicle Management**
3. **Trip & Dispatch Management**
4. **Maintenance & Fuel Management**
5. **Document, Compliance & Admin Management**

4.1.1 USER & DRIVER MANAGEMENT

The User & Driver Management module allows administrators to manage user accounts and driver profiles. It stores essential personal, contact, licence, and role-related information. Role-based access ensures that each user can access only the functions relevant to their responsibilities.

Table 4.1 User & Driver Management

Field	Description
User ID	Unique identification number for each user
Name	Full name of the user or driver
Email	Email address used for login and communication
Contact Number	Phone number of the user or driver
Role	Defines the user's responsibility, such as Admin, Fleet Manager, or Driver
Licence Number	Driving licence number of the driver
Licence Expiry	Expiry date of the driving licence

4.1.2 VEHICLE MANAGEMENT

The Vehicle Management module maintains complete information about fleet vehicles. It allows administrators to add, update, and monitor vehicles along with their current status and assigned drivers. This helps in efficient vehicle utilization and tracking.

Table 4.2 Vehicle Management

Field	Description
Vehicle ID	Unique identification number for each vehicle
Vehicle Number	Registration number of the vehicle
Vehicle Type	Type of vehicle such as Car, Van, Bus, or Truck
Model	Vehicle model and variant
Status	Current status such as Available, In Use, or Under Maintenance
Assigned Driver	Driver assigned to the vehicle
Registration Date	Date on which the vehicle was registered
Insurance Expiry	Expiry date of vehicle insurance

4.1.3 TRIP & DISPATCH MANAGEMENT

The Trip & Dispatch Management module manages trip planning, scheduling, and assignment of vehicles and drivers. It records trip details from the starting point to the destination and helps fleet managers monitor ongoing and completed trips.

Table 4.3 Trip & Dispatch Management

Field	Description
Trip ID	Unique identification number for each trip
Vehicle ID	Vehicle assigned for the trip
Driver ID	Driver assigned to the trip
Source	Starting location of the trip
Destination	Final destination of the trip

Start Date & Time	Scheduled or actual trip starting time
Expected Arrival	Expected arrival date and time
Distance	Total planned or travelled distance
Trip Status	Status such as Scheduled, Ongoing, Completed, or Cancelled

4.1.4 MAINTENANCE & FUEL MANAGEMENT

The Maintenance & Fuel Management module manages vehicle servicing, maintenance records, and fuel consumption. It helps schedule preventive maintenance and monitor fuel expenses, improving vehicle reliability and controlling operational costs.

Table 4.4 Maintenance & Fuel Management

Field	**Description**
Maintenance ID	Unique identification number for maintenance records
Vehicle ID	Vehicle associated with the maintenance record
Service Type	Type of service or maintenance performed
Service Date	Date on which the vehicle was serviced
Fuel Record ID	Unique identification number for fuel records
Fuel Quantity	Amount of fuel consumed
Fuel Cost	Cost incurred for fuel
Maintenance Status	Current maintenance status of the vehicle

4.1.5 DOCUMENT, COMPLIANCE & ADMIN MANAGEMENT

The Document, Compliance & Admin Management module manages important vehicle and driver documents and monitors their validity. It provides expiry alerts and compliance information. The admin dashboard gives a centralized view of fleet activities, alerts, and reports for effective decision-making.

Table 4.5 Document, Compliance & Admin Management

Field	Description
Document ID	Unique identification number for each document
Vehicle/Driver ID	Vehicle or driver associated with the document
Document Type	Type of document such as Registration, Insurance, Pollution, Fitness, or Licence
Issue Date	Date on which the document was issued
Expiry Date	Date on which the document expires
Document Status	Current status such as Valid, Expired, or Pending
Alert Status	Indicates whether an expiry notification has been generated
Reports & Analytics	Provides fleet performance and operational reports

4.2 USE CASE DIAGRAM

A use case diagram is a visual representation in UML (Unified Modelling Language) that illustrates the interactions between actors and a system or software application.

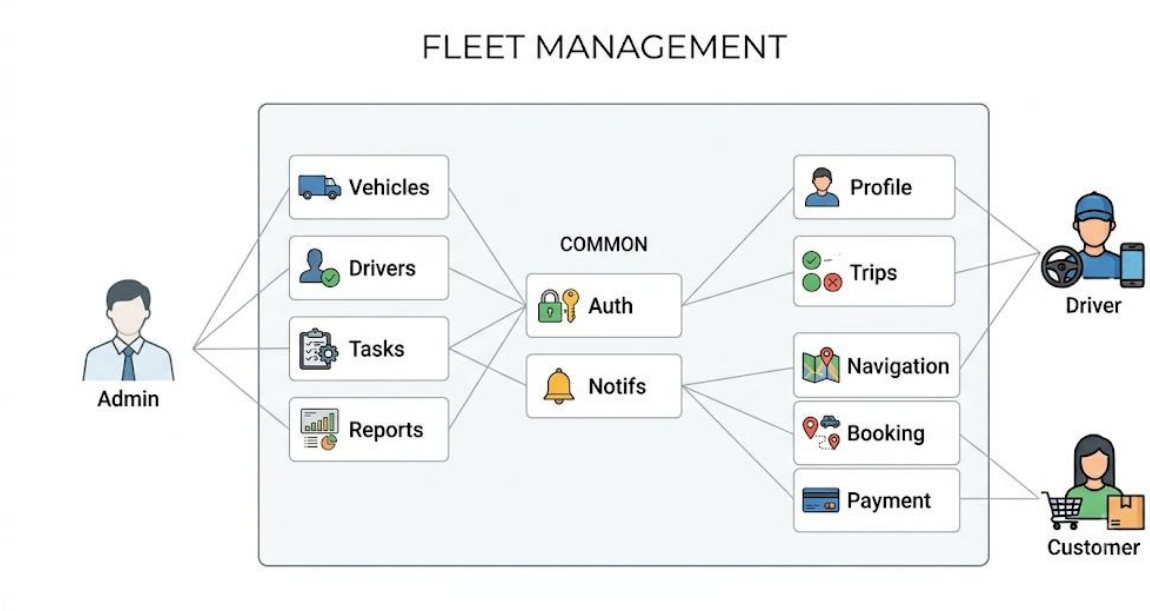


Fig. 4.1 Use Case Diagram

4.3 SEQUENCE DIAGRAM

A sequence diagram is a visual representation used in software engineering to illustrate the interactions and communication between different objects or components in a system over a specific period of time. It shows the chronological order of messages or method calls exchanged between these entities, helping to depict the dynamic behaviour of a system or a particular scenario. In essence, it provides a time-ordered view of how various parts of a system collaborate to achieve a particular task or functionality.

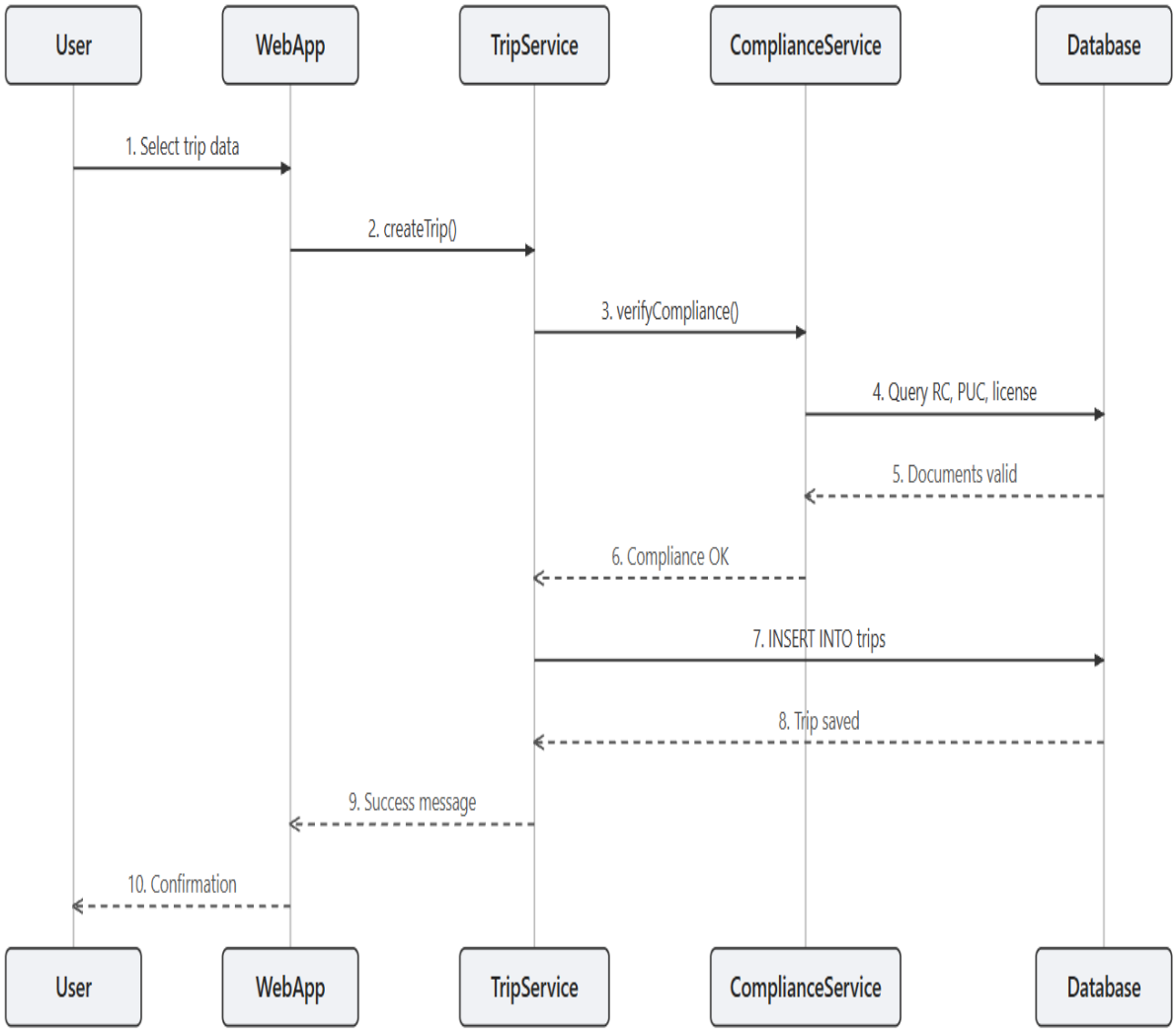


Fig. 4.2 Sequence Diagram

4.4 DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is a visual representation that illustrates the flow of data within a system or process. It uses symbols to depict processes, data stores, data flow, and external entities. DFD helps to analyse, design, and document information systems, showing how data is input, processed and output, while emphasizing the interactions between different elements in the system.

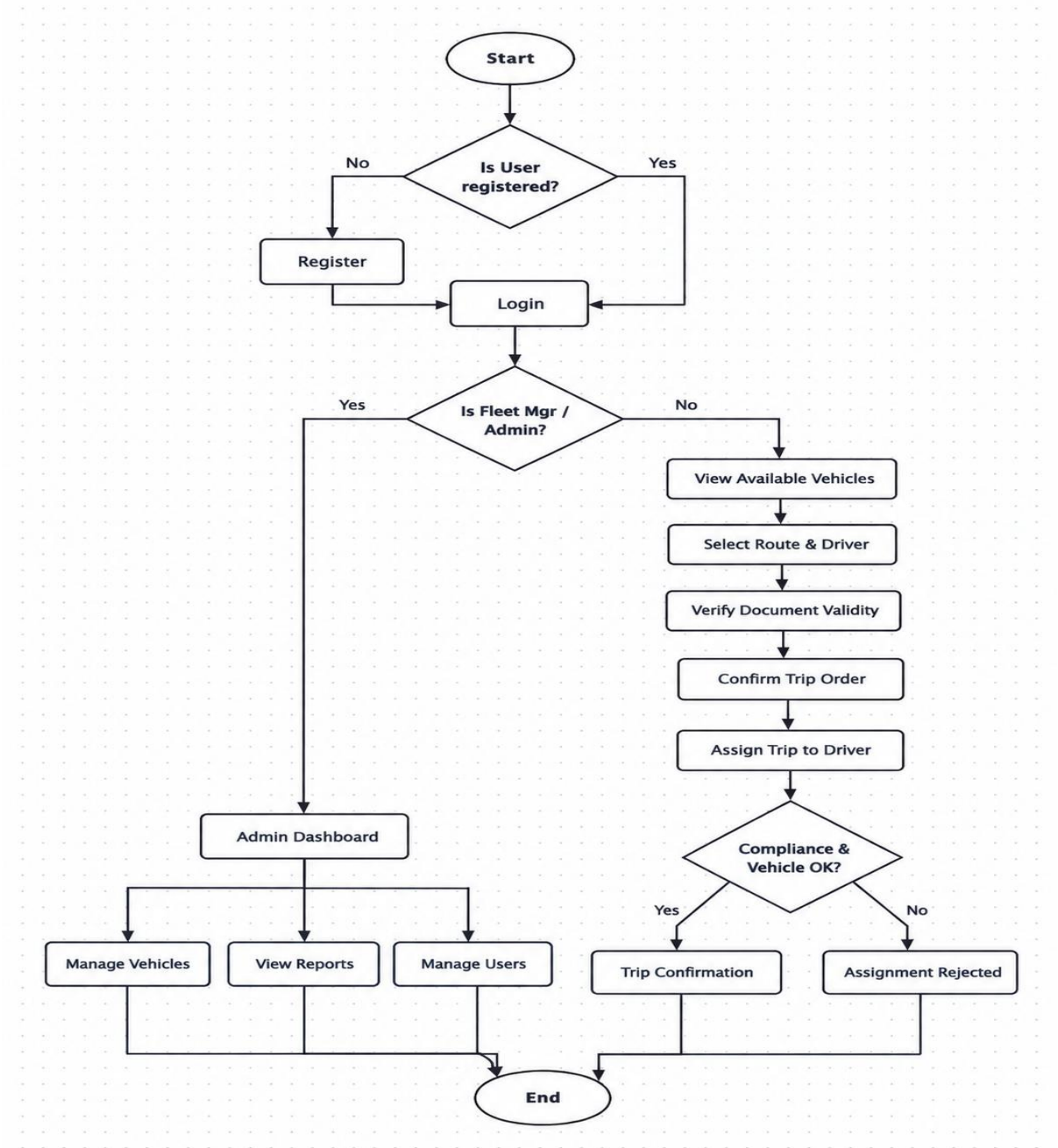


Fig. 4.3 Data Flow Diagram

CHAPTERN – 5

TESTING

5.1 UNIT TESTING

Unit testing forms the foundational layer of the testing process for the Fleet Management System. It involves testing individual software components in isolation to identify and rectify issues early in the development cycle. In this context, unit testing includes examining Java classes, service methods, validation logic, and database queries individually. Unit testing, often conducted concurrently with development, focuses on validating the smallest units of the software and ensuring that each component works correctly before being integrated into the complete system.

5.2 INTEGRATION TESTING

Integration testing for the Fleet Management System focuses on ensuring seamless interaction between different components of the system. The goal is to validate that API endpoints responsible for communication between the React front-end and Java Spring Boot back-end work correctly together. This phase ensures accurate data flow between modules such as vehicle management, driver management, trip management, maintenance, fuel, and document management. It ultimately validates the overall functionality and coordination of the fleet management system.

5.3 SECURITY AND AUTHENTICATION

In the Fleet Management System, ensuring the security and authentication of user data is of utmost importance. JSON Web Tokens (JWTs) are used for user authentication and secure access control. During user registration, user credentials are securely stored in the database. After successful login, a JWT is generated and issued to the authenticated

user. The JWT serves as an authentication token, allowing secure access to the system's features according to the user's assigned role.

The JWT is securely signed using a secret key known only to the server, ensuring the authenticity and integrity of the token. Whenever a user interacts with the system, the JWT is included in the request. The Spring Boot backend validates the token before granting access to protected resources. Role-based authorization ensures that administrators, fleet managers, and drivers can access only the functions permitted for their responsibilities. To enhance security, JWTs have a finite expiration time, after which the user must authenticate again to obtain a new token.

```
const handleSelectDemo = (key) => {
    const demo = DEMO_ACCOUNTS[key];
    if (demo) {
      setIdentifier(demo.email);
      setPassword(demo.password);
      setSelectedDemo(key);
    }
  };

  const handleLogin = async (e) => {
    e?.preventDefault();
    if (!identifier.trim() || !password.trim()) {
      toast.warning('Please enter your identifier and password')
      return;
    }

    setLoading(true);
    try {
      const authData = await login({
        identifier: identifier.trim(),
        email: identifier.trim(),
        password
      });
      toast.success(`Welcome back, ${authData.fullName}!`);

      // Role-based auto navigation
      if (authData.role === 'ADMIN') navigate('/admin/dashboard'
      else if (authData.role === 'DRIVER') navigate(
'/driver/dashboard');
      else if (authData.role === 'CUSTOMER') navigate(
'/customer/dashboard');
      else navigate('/');
    } catch (err) {
      const status = err.response?.status;
      const responseData = err.response?.data;

      if (!err.response) {
        toast.error(
'Unable to connect to the server. Please try again.');
      } else if (status === 401) {
        toast.error(responseData?.message ||
'Invalid email or password.');
      } else if (status === 500) {
        toast.error(
'Something went wrong on the server. Please check the backend lo
);
      } else if (responseData?.validationErrors && typeof
responseData.validationErrors === 'object') {
        const specificMsg = Object.values(responseData
.validationErrors).join('. ');
        toast.error(specificMsg || responseData.message ||
'Validation error');
      } else {
        toast.error(responseData?.message ||
'Authentication failed. Please check your credentials.');
      }
    } finally {
      setLoading(false);
    }
  };
```

Fig. 5.1. Handling User Authentication and Role-Based Redirection

```
// Attach JWT token to all outgoing requests
api.interceptors.request.use(
  (config) => {
    const token = localStorage.getItem('fleet_token');
    if (token) {
      config.headers.Authorization = `Bearer ${token}`;
    }
    return config;
  },
  (error) => Promise.reject(error)
);
```

Fig. 5.2 Authenticating the User using Bearer Token

5.2 TEST CASES

Test cases are essential for evaluating the functionality of the **Fleet Management System**, including user authentication, vehicle management, driver management, trip scheduling, vehicle assignment, and fleet tracking. Proper testing ensures data accuracy, secure access, smooth operation, and a seamless user experience while maintaining the overall quality, reliability, and efficiency of the system.

5.4.1 TEST CASE I

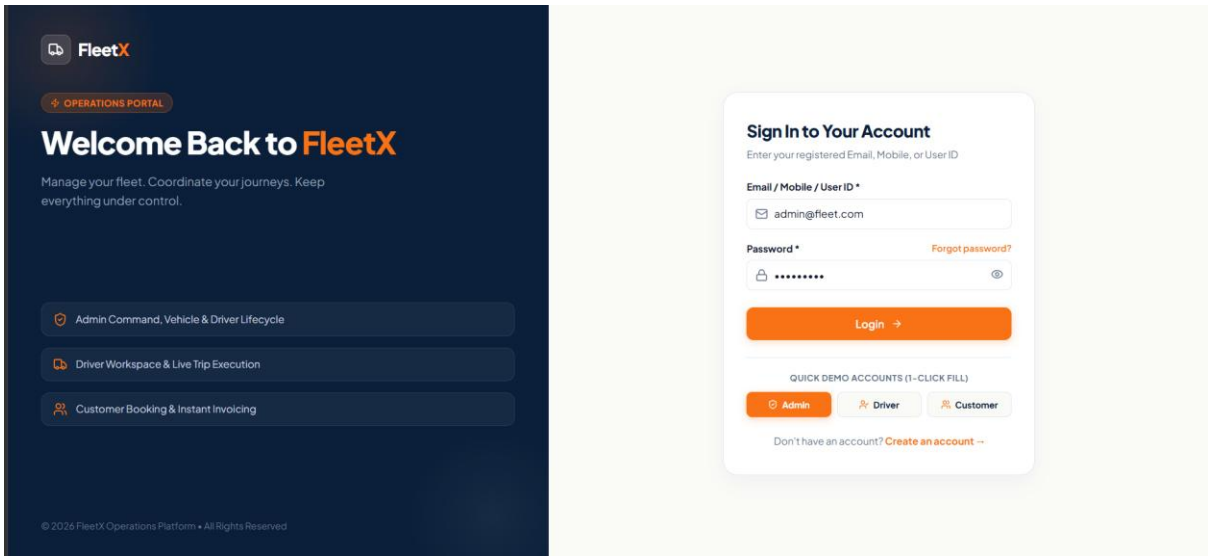


Fig. 5.3 Test Case I

EXPECTED OUTPUT:

Registration fails and an error message is displayed, prompting the user to enter the missing required credentials.

ACTUAL OUTPUT:

Registration fails, and the system indicates that all required credentials must be entered.

5.4.2 TEST CASE II

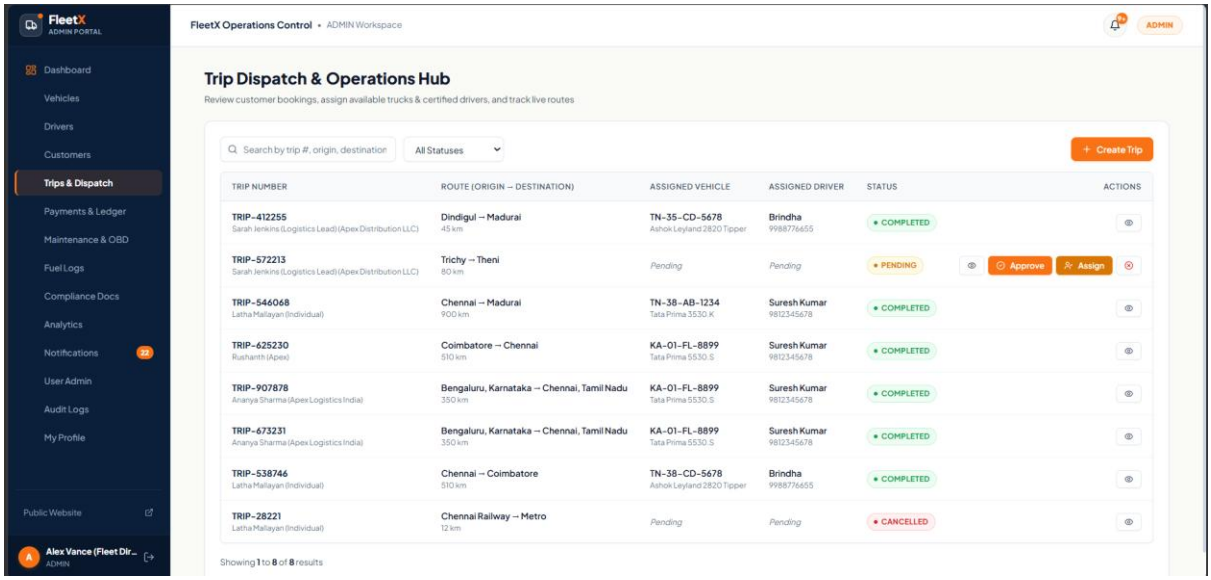


Fig. 5.4 Test Case II

EXPECTED OUTPUT:

After successfully booking a vehicle, a confirmation message is displayed indicating that the booking was completed successfully.

ACTUAL OUTPUT:

After booking the vehicle, the system displays a **"Trip Completed successfully"** message.

CHAPTER – 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The Fleet Management System project is an essential solution aimed at addressing the growing need for efficient and organized fleet operations. By providing a centralized platform for managing vehicles, drivers, trips, maintenance, fuel, and compliance records, the system improves operational efficiency while reducing manual effort. Its focus on centralized data management, preventive maintenance, fuel monitoring, document tracking, and secure role-based access ensures a reliable and efficient fleet management experience. Overall, this project represents a significant step toward modernizing fleet operations, reducing operational errors and costs, improving vehicle utilization, and supporting better decision-making for fleet managers.

6.2 FUTURE WORK

An alternative solution for enhancing fleet management could be to develop a mobile application that allows fleet managers and drivers to manage important fleet activities from a single platform. The application could include features such as real-time vehicle tracking, trip updates, maintenance reminders, fuel monitoring, driver notifications, and digital document access. This approach would provide greater convenience and accessibility to users while enabling fleet operations to be monitored from anywhere.

For fleet administrators and managers, an advanced online dashboard could be developed to provide real-time fleet monitoring and intelligent analytics. The dashboard could include features such as predictive maintenance, GPS-based vehicle tracking, fuel efficiency analysis, driver performance monitoring, demand forecasting, automated compliance alerts, and detailed fleet utilization reports. Such enhancements would provide a centralized solution for fleet operations, improving efficiency, reducing operational costs, increasing vehicle utilization, and supporting data-driven decision-making.

CHAPTER – 7

APPENDICEES

APPENDIX I SOURCE CODE

```
import api from './api';

export const tripService = {
  getTrips: async (params = {}) => {
    const response = await api.get('/trips', { params });
    return response.data;
  },

  getTripById: async (id) => {
    const response = await api.get(`/trips/${id}`);
    return response.data;
  },

  requestTrip: async (tripData) => {
    const response = await api.post('/trips', tripData);
    return response.data;
  },

  approveTrip: async (id) => {
    const response = await api.post(`/trips/${id}/approve`
);
    return response.data;
  },

  assignTrip: async (id, assignmentData) => {
    const response = await api.post(`/trips/${id}/assign`,
assignmentData);
    return response.data;
  },

  startTrip: async (id) => {
    const response = await api.post(`/trips/${id}/start`);
    return response.data;
  },
```

```
  completeTrip: async (id, data = {}) => {
    const response = await api.post(`/trips/${id}/complete`
, data);
    return response.data;
  },

  cancelTrip: async (id, reasonData = {}) => {
    const response = await api.post(`/trips/${id}/cancel`,
reasonData);
    return response.data;
  },

  // Role-specific helpers
  getMyDriverTrips: async () => {
    const response = await api.get('/trips/driver/my-trips'
);
    return response.data;
  },

  getMyDriverActiveTrips: async () => {
    const response = await api.get('/trips/driver/active');
    return response.data;
  },

  getMyCustomerTrips: async () => {
    const response = await api.get(
'/trips/customer/my-trips');
    return response.data;
  },

  getMyCustomerActiveTrips: async () => {
    const response = await api.get('/trips/customer/active'
);
    return response.data;
  }
};
```

APPENDIX II SCREENSHOTS

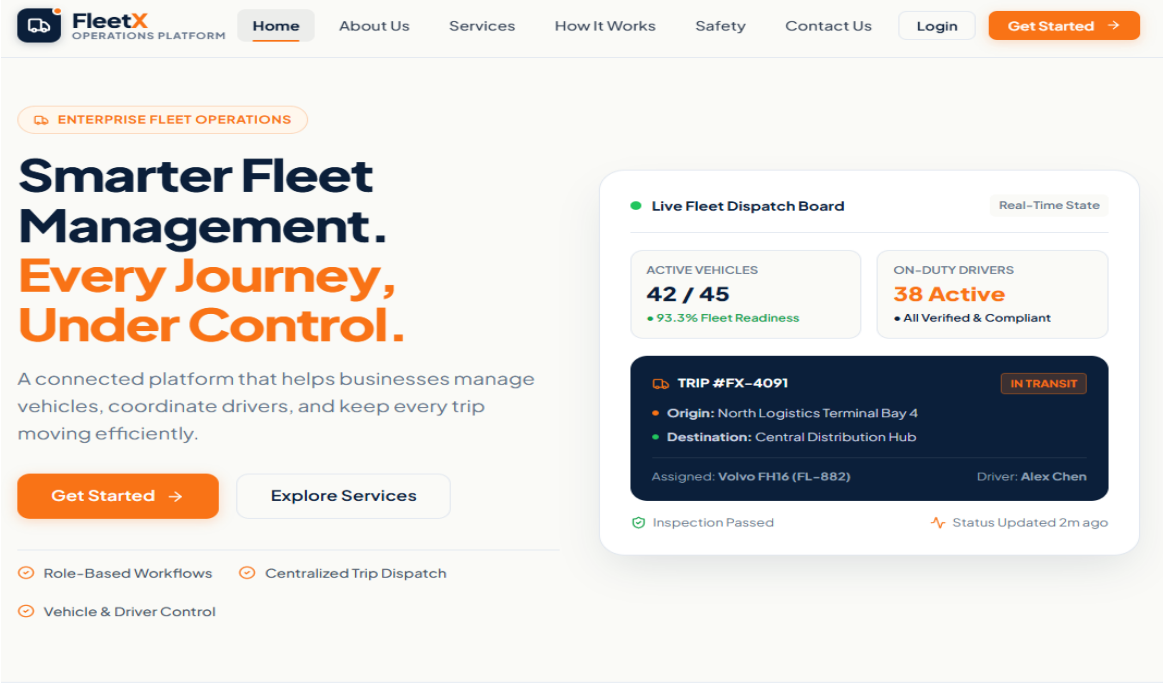


Fig. A.2.1 Home Page

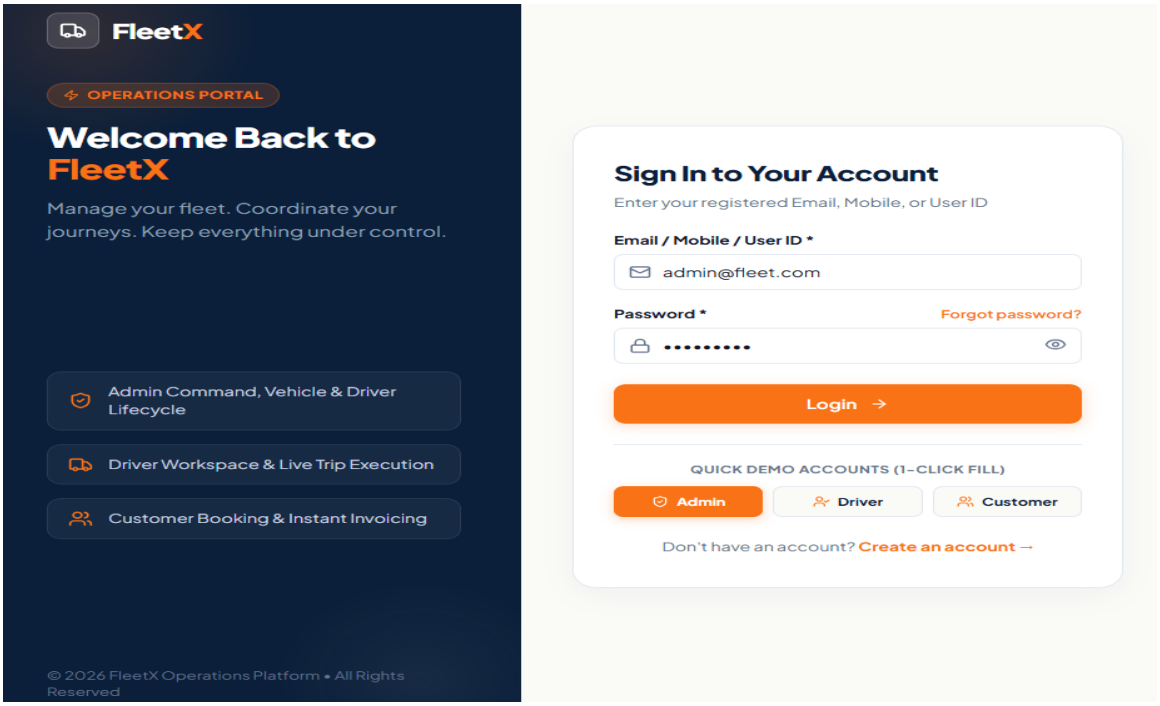


Fig. A.2.2 Admin Login

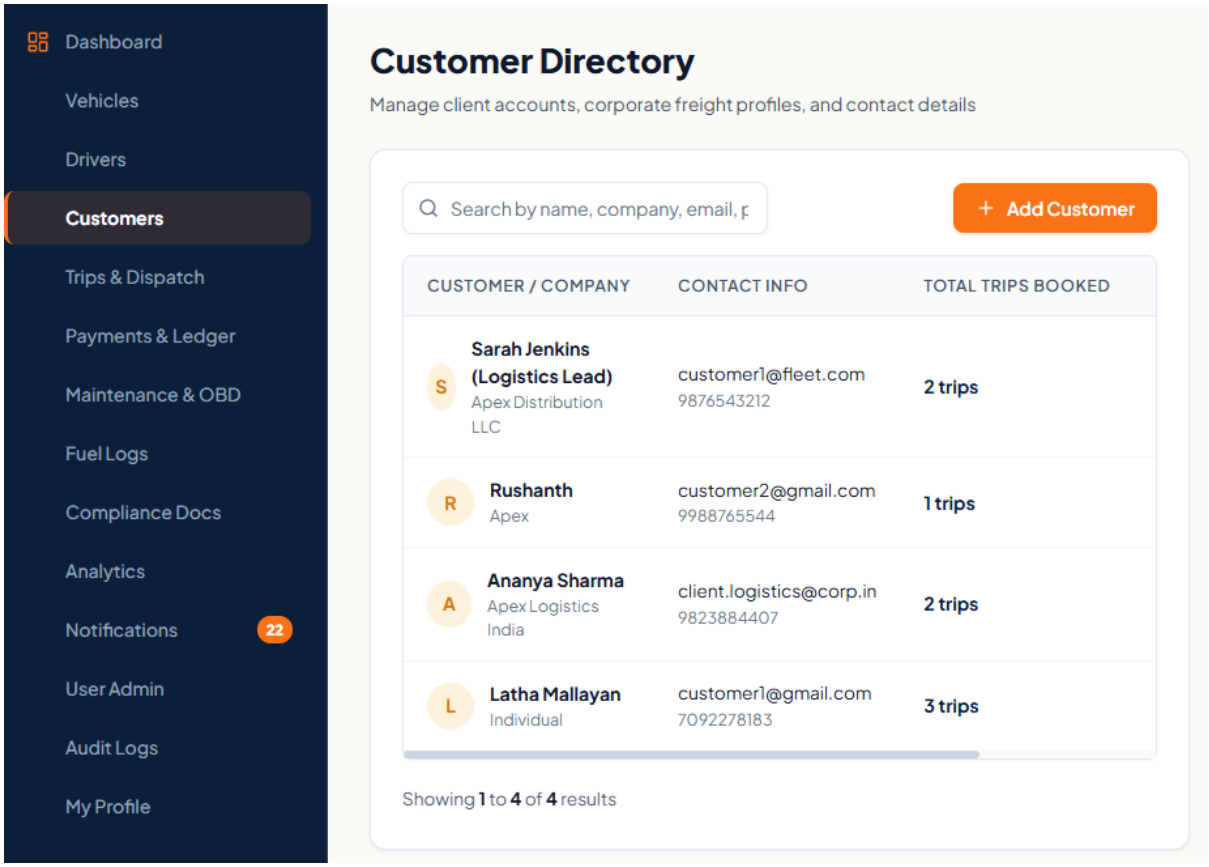


Fig. A.2.3 Customer Management

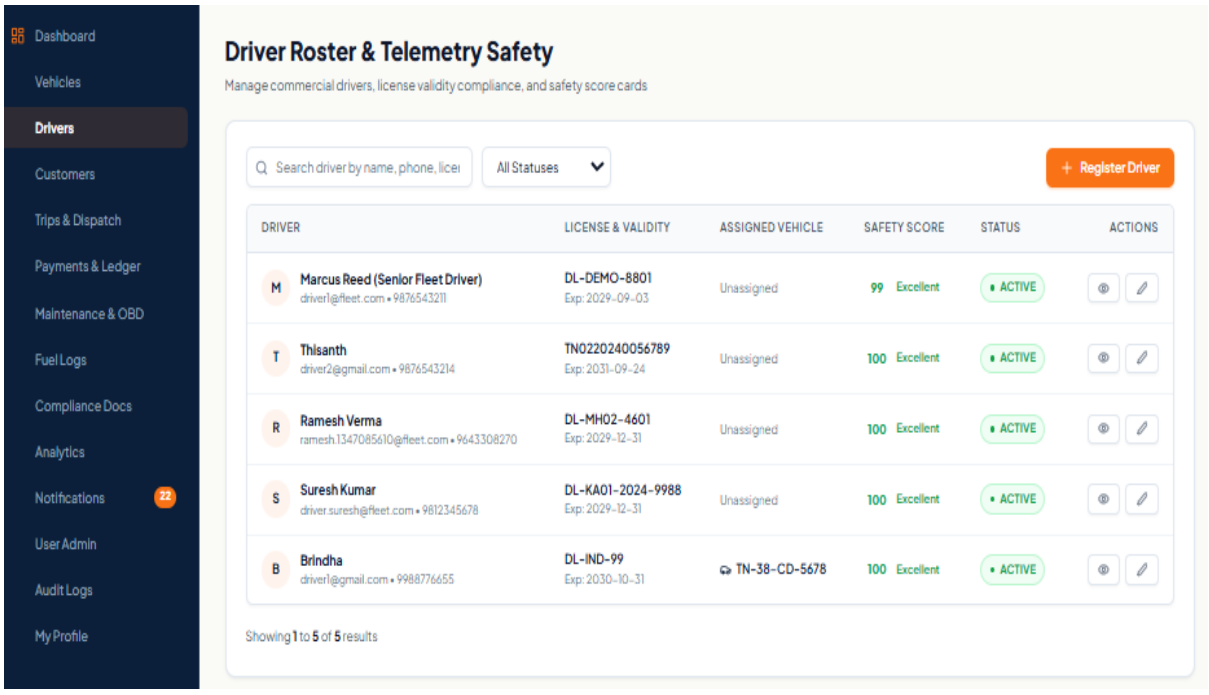


Fig. A.2.4 Driver Management

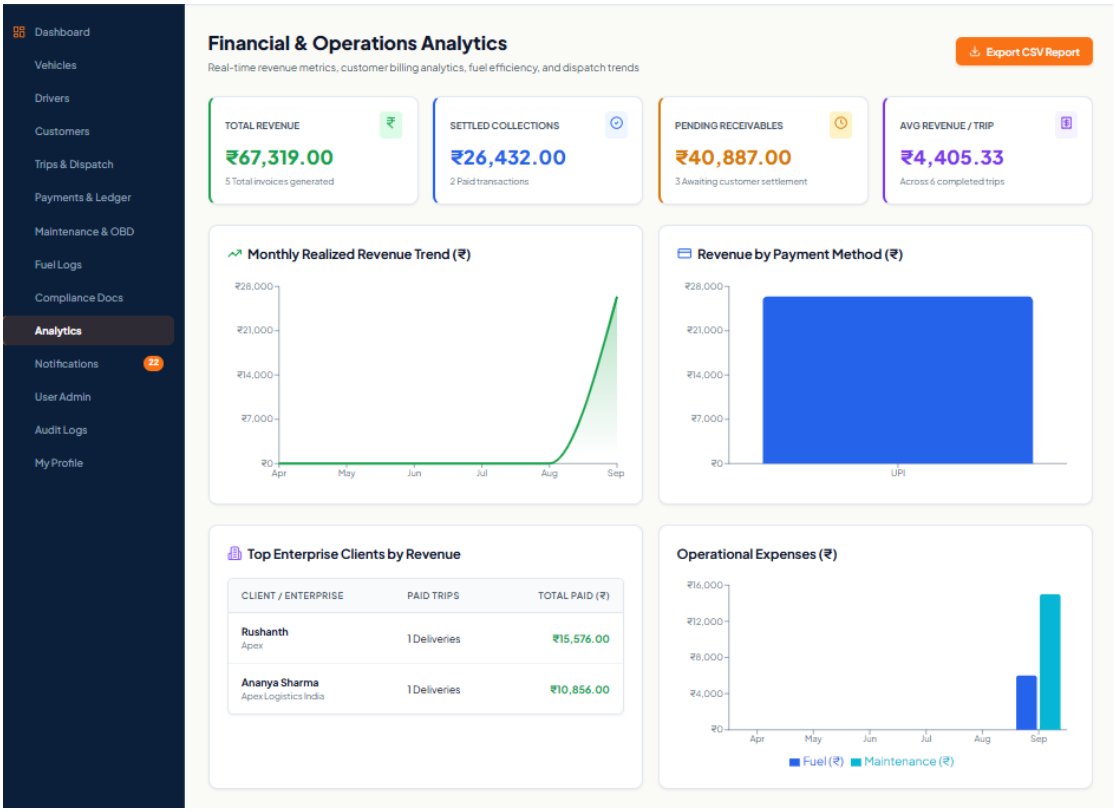


Fig. A.2.5 Analytics

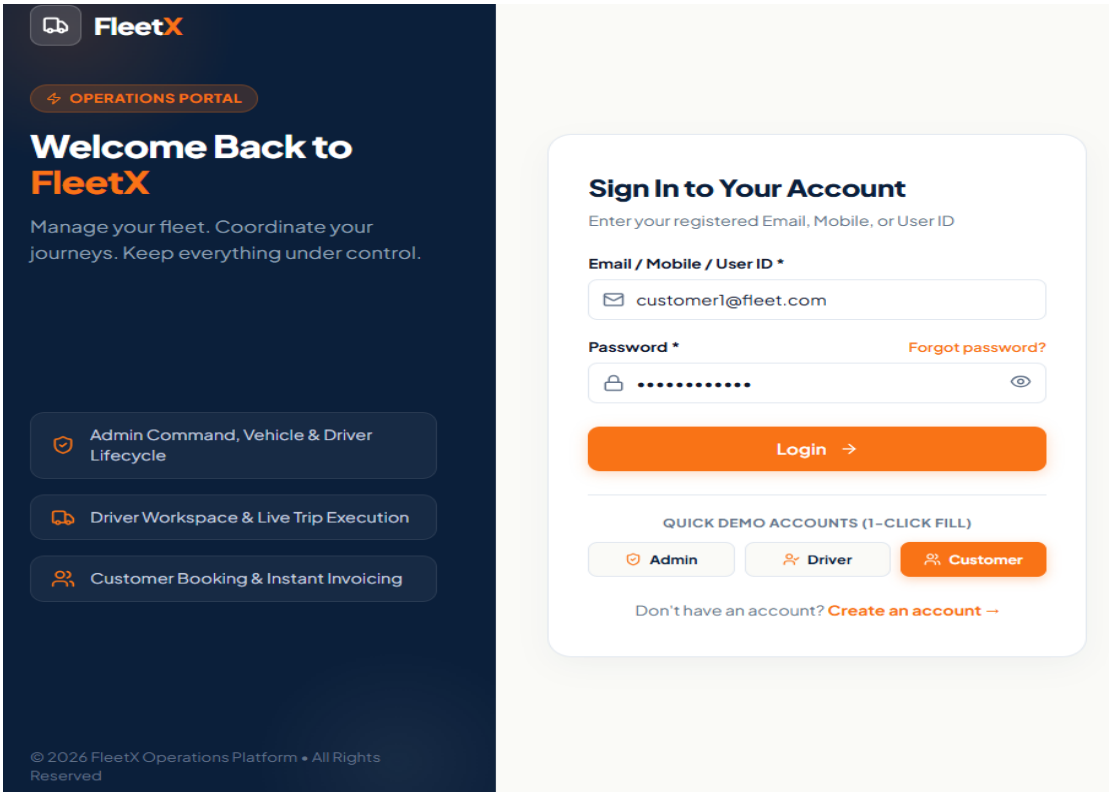


Fig. A.2.6 Customer Login

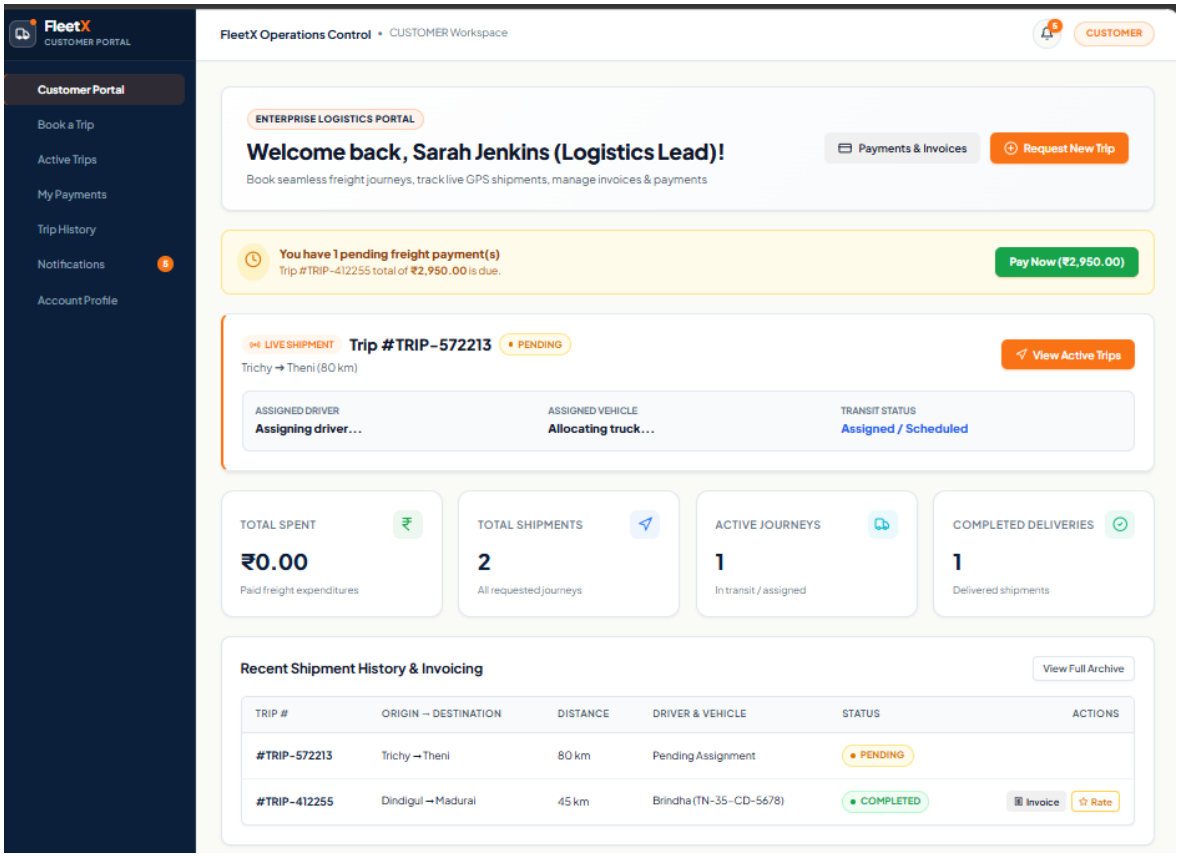


Fig. A.2.7 Customer Dashboard

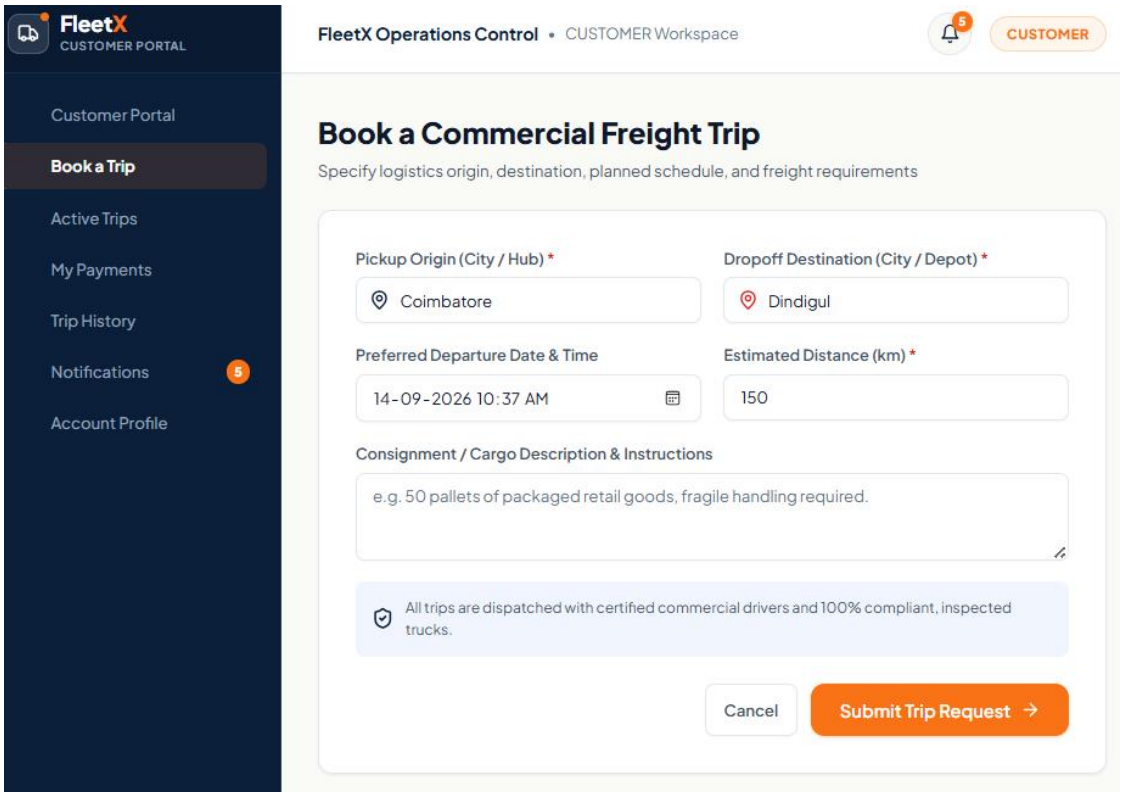


Fig. A.2.8 Book a Trip

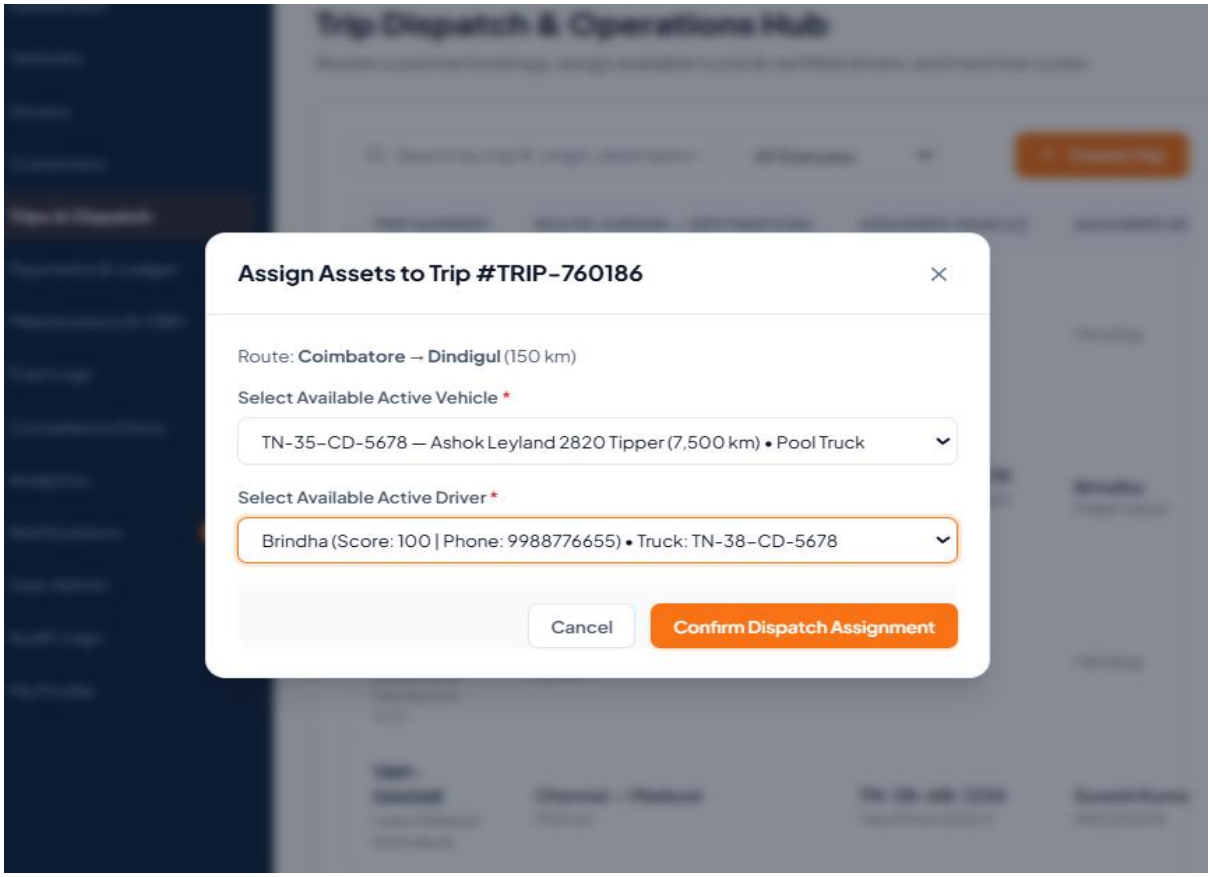


Fig. A.2.9 Assign Assets to Trip

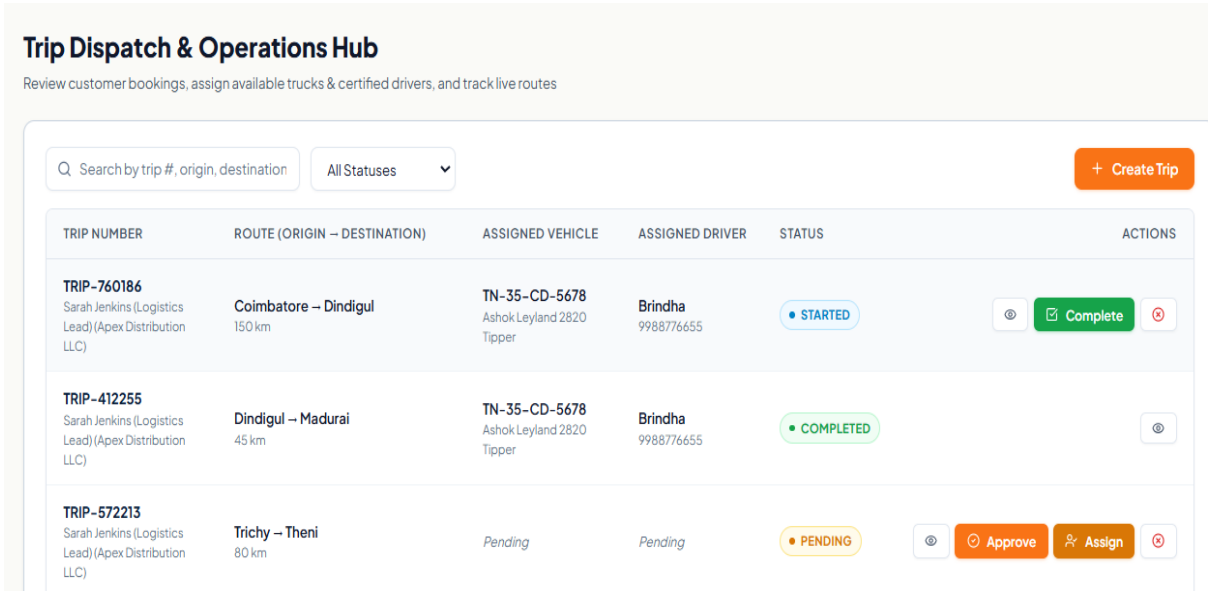


Fig. A.2.10 Completion Status

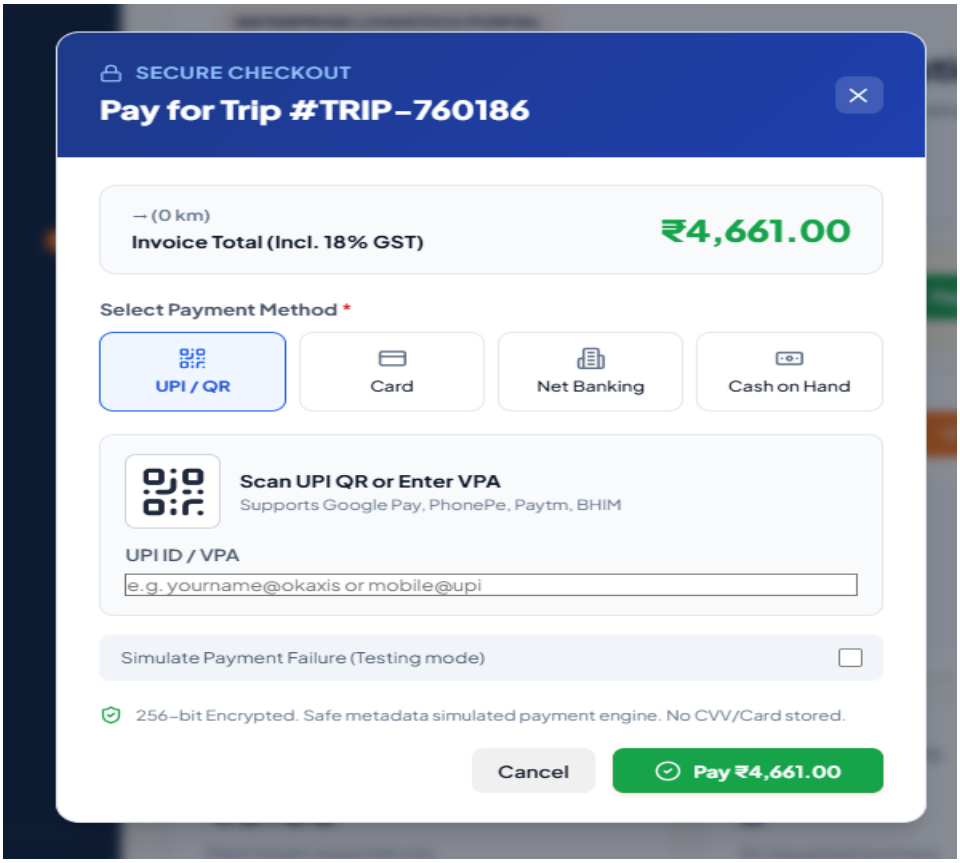


Fig. A.2.11 Payment Method Details

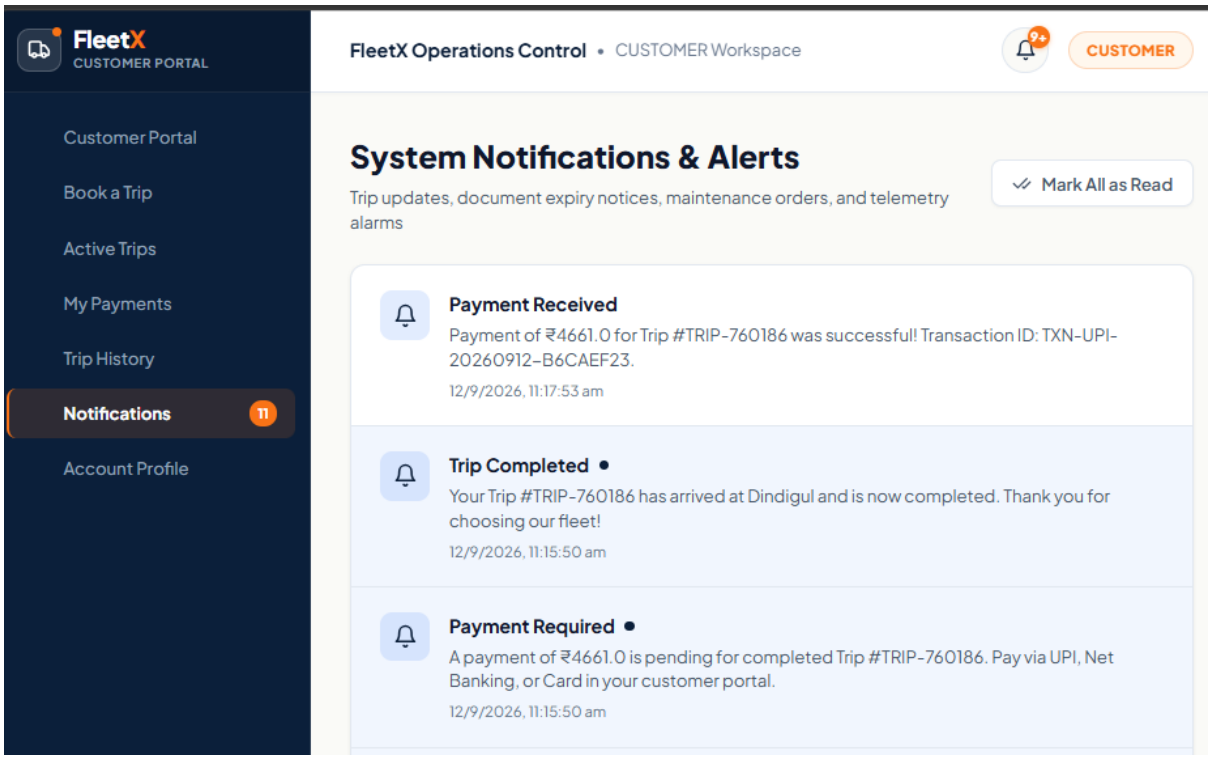
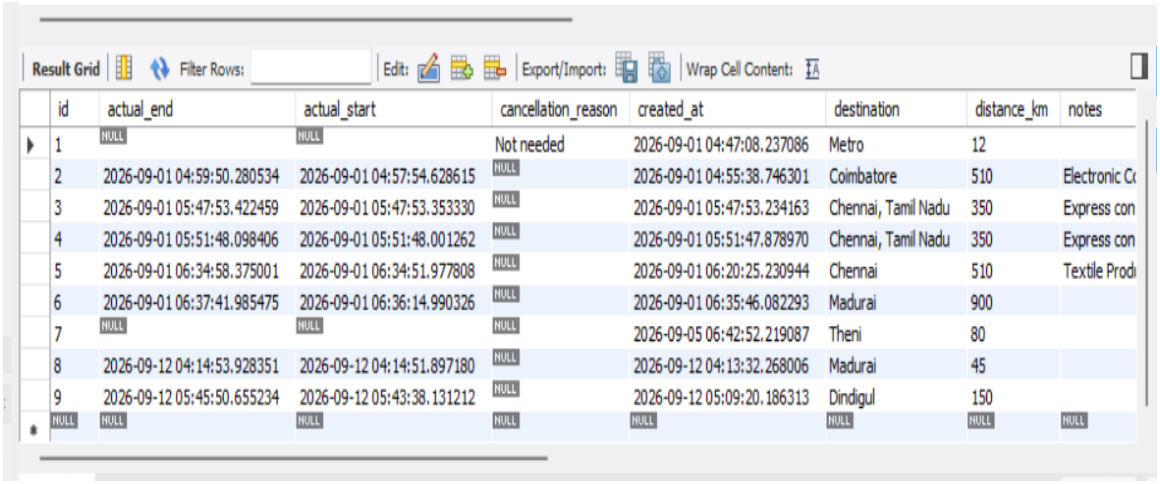


Fig. A.2.12 Overall Status



Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	actual_end	actual_start	cancellation_reason	created_at	destination	distance_km	notes
1	NULL	NULL	Not needed	2026-09-01 04:47:08.237086	Metro	12	
2	2026-09-01 04:59:50.280534	2026-09-01 04:57:54.628615	NULL	2026-09-01 04:55:38.746301	Coimbatore	510	Electronic C
3	2026-09-01 05:47:53.422459	2026-09-01 05:47:53.353330	NULL	2026-09-01 05:47:53.234163	Chennai, Tamil Nadu	350	Express con
4	2026-09-01 05:51:48.098406	2026-09-01 05:51:48.001262	NULL	2026-09-01 05:51:47.878970	Chennai, Tamil Nadu	350	Express con
5	2026-09-01 06:34:58.375001	2026-09-01 06:34:51.977808	NULL	2026-09-01 06:20:25.230944	Chennai	510	Textile Prod
6	2026-09-01 06:37:41.985475	2026-09-01 06:36:14.990326	NULL	2026-09-01 06:35:46.082293	Madurai	900	
7	NULL	NULL	NULL	2026-09-05 06:42:52.219087	Theni	80	
8	2026-09-12 04:14:53.928351	2026-09-12 04:14:51.897180	NULL	2026-09-12 04:13:32.268006	Madurai	45	
9	2026-09-12 05:45:50.655234	2026-09-12 05:43:38.131212	NULL	2026-09-12 05:09:20.186313	Dindigul	150	
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Fig. A.2.13 Trip DataBase

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